

CHAPTER 3 - Numerical Descriptive Measures

SDS188 - Group 3

2026

1 Notation and rules

This notation is helpful especially when the sample size is very large!

General:

Let X = variable of interest

n = size of sample/data

i = index for the observation, where $i = 1, 2, 3, \dots, n$

x_1 is the first observation

x_2 is the second observation

$\vdots$

x_i is the i^{th} observation

$\vdots$

x_i is the n^{th} observation

1.1 Example 1

(Data vanaf Frekwensieverdeling Example in Chapter 2)

Data: daily maximum temperature

24	35	17	21	24
37	26	46	58	30
32	13	12	38	41
43	44	27	53	27

We can rewrite this data in a different notation

Let X = daily maximum temperature

$n = 20$

$i = 1, 2, \dots, 20$

Index (i)	Temperature (X)
1	24
2	35
3	17
$\vdots$	$\vdots$
18	27
19	53
20	27

1.1.1 The sum or total of observed values

$$\sum X = \sum_{i=1}^n x_i$$

$$24 + 35 + 17 + 21 + \dots + 44 + 27 + 53 + 24 = 648$$

OF

$$\sum_{i=1}^n x_i = x_1 + x_2 + x_3 + \dots + x_{n-2} + x_{n-1} + x_n$$

$$\sum_{i=1}^{20} x_i = x_1 + x_2 + x_3 + \dots + x_{18} + x_{19} + x_{20}$$

$$= 24 + 35 + 17 + \dots + 27 + 53 + 27$$

$$= 648$$

CASIO calculator steps:

1. MODE -> 2:STAT -> 1:1-VAR
2. 24 = 35 = 17 = ... = 27 = 53 = 27
3. AC
4. SHIFT -> 1 -> 3:Sum -> 2: $\sum x$
5. =

$$\sum_{i=1}^{20} = 648$$

1.1.2 The sum or total of the squared observed values

$$\sum X = \sum_{i=1}^n x_i^2$$

$$\sum_{i=1}^n x_i^2 = x_1^2 + x_2^2 + x_3^2 + \cdots + x_{n-2}^2 + x_{n-1}^2 + x_n^2$$

$$\sum_{i=1}^{20} x_i^2 = x_1^2 + x_2^2 + x_3^2 + \cdots + x_{18}^2 + x_{19}^2 + x_{20}^2$$

$$= 24^2 + 35^2 + 17^2 + \cdots + 27^2 + 53^2 + 27^2$$

$$= 576 + 1225 + 289 + \cdots + 729 + 2809 + 729$$

$$= 24046$$

CASIO calculator steps:

1. MODE -> 2:STAT -> 1:1-VAR
2. 24 = 35 = 17 = ... = 27 = 53 = 27
3. AC
4. SHIFT -> 1 -> 3:Sum -> 1: $\sum x^2$
5. =

$$\sum_{i=1}^{20} = 24046$$

1.2 Example 2 - Two variables

Data: information regarding newborn babies

Let X = length of baby at birth (cm)

Y = weight of baby at birth (kg)

$$n = 42$$

$$i = 1, 2, \dots, 42$$

i	X	Y
1	56	4.55
2	53	4.32
3	58	4.1
4	53	4.07
5	48	3.11
6	51	3.93
7	52	3.77
8	50	3.47
9	49	3.32
10	58	4.57

CASIO steps for inputting data:

1. MODE -> 2:STAT -> 2:A+BX
2. 56 = 53 = 58 = ... = 50 = 49 = 58
3. ► ▲
4. 4.55 = 4.32 = 4.1 = ... = 3.47 = 3.32 = 4.57
5. AC

QUESTIONS:

1. $\sum_{i=1}^{10} x_i =$ [SHIFT -> 1 -> 3:Sum -> 2: $\sum x$]
2. $\sum_{i=1}^{10} y_i =$ [SHIFT -> 1 -> 3:Sum -> 4: $\sum y$]
3. $\sum_{i=1}^{10} x_i y_i =$ [SHIFT -> 1 -> 3:Sum -> 5: $\sum xy$]

1.3 Example 3 - Constants

1.3.1 Example 3.1

Let $c = 5$:

Data: 39, 29, 43, 52, 39, 44, 40, 31, 44, 35

i	X	cX
1	39	$5 \times 39 = 195$
2	29	$5 \times 29 = 145$
3	43	$5 \times 43 = 215$
4	52	$5 \times 52 = 260$
5	39	$5 \times 39 = 195$
6	44	$5 \times 44 = 220$
7	40	$5 \times 40 = 200$
8	31	$5 \times 31 = 155$
9	44	$5 \times 44 = 220$
10	35	$5 \times 35 = 175$
Total	$\sum_{i=1}^{10} x_i = 396$	$\sum_{i=1}^{10} 5x_i = 1980$

$$\begin{aligned}
\sum cX &= \sum_{i=1}^{10} 5x_i \\
&= (5 \times 39) + \cdots + (5 \times 35) \\
&= 195 + \cdots + 175 \\
&= 1980 \\
5 \sum_{i=1}^{10} &= 5 \times 396 \\
&= 1980
\end{aligned}$$

1.3.2 Example 3.2

Let $c = 5$ and $n = 10$

$i = 1, 2, \dots, 10$

$\sum_{i=1}^{10} 5(x_i)$ where $x_i = 1$

$$\begin{aligned}
\sum_{i=1}^{10} 5x_i &= 5x_1 + 5x_2 + \cdots + 5x_9 + 5x_{10} \\
&= 5(1) + 5(1) + \cdots + 5(1) + 5(1) \\
&= 5 + 5 + \cdots + 5 + 5 \\
&= 10 \times 5 \\
&= 50
\end{aligned}$$

$$\therefore \sum_{i=1}^{10} 5 = n \times 5 = 10 \times 5 = 50$$

1.4 Applications and rules

- $\sum X^2 \neq (\sum X)^2$
- $\sum XY \neq (\sum X)(\sum Y)$
- $\sum X + Y \neq \sum X + \sum Y$
- $\sum X - Y \neq \sum X - \sum Y$
- $\sum cX \neq c(\sum X)$
- $\sum c \neq nc$

2 Measures of central tendency

Central tendency: where is the most of the data located/ where is the data concentrated

2.1 The mean

$$\bar{x}$$

Excel: =AVERAGE(data)

CASIO: SHIFT -> 1 -> 4:Var -> 2: $\bar{x}$

Formulas:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

2.1.1 Population mean

$$\mu$$

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

2.2 The median

$$Q_2$$

Excel: =MEDIAN(data)

1. Sort data in ascending order
2. Determine position of the median with $\frac{1}{2}(n + 1)$
3. Find the median as the value in the $\frac{1}{2}(n + 1)$ th position

If n is an even number -> The median is the value in this position, for example $n = 12$, median position = $\frac{1}{2}(12) = 6$.

If n is an uneven number -> The median is the average of the values, for example $n = 15$, median position = $\frac{1}{2}(15) = 7.5$, the median is the average of the 7th and the 8th values.

2.3 The mode

Excel: =MODE(data)

The value that occurs the most

- Sometimes the data has more than one mode
- Sometimes the data has no mode
- Compare with the mean and median to determine symmetry

2.4 The geometric mean

$$\bar{x}_G$$

Excel: =GEOMEAN(data)

A great advantage of the geometric mean is that the amounts that are invested do not have to be known; the calculation focuses on the returns and provides a method to compare two investment options over time.

The geometric mean measures the change over time and will always be smaller than $\bar{x}$.

2.4.1 The geometric mean rate of return

$$\bar{r}_G$$

The geometric mean rate of return ($\bar{r}_G$) measures the status of the investment over time.

$$r_i = \frac{x_i - x_{i-1}}{x_{i-1}}$$

$$\bar{r}_G = \left[\prod_{i=1}^n (1 + r_i) \right]^{\frac{1}{n}} - 1$$

3 Measures of variation

Variation: how the data is spread around the measure of central tendency

3.1 The range

$$X_{largest} - X_{smallest}$$

OF

$$Maximum - Minimum$$

Excel:

$X_{largest}$ OF *Maximum* =MAX(data)

$X_{smallest}$ OF *Minimum* =MIN(data)

CASIO:

$X_{largest}$ OF *Maximum*: SHIFT -> 1 -> 5:MinMax -> 1:maxX

$X_{smallest}$ OF *Minimum*: SHIFT -> 1 -> 5:MinMax -> 2:minX

Interpretation: The largest difference in *X* for any two *items* is *variasiewydte*.

3.2 The variance and standard deviation

3.2.1 Formulas for variance and standard deviation

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$s^2 = \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2 \right]$$

$$s = \sqrt{s^2}$$

3.2.2 Calculation of variance - method 1

i	X	$X_i - \bar{X}$	$(X_i - \bar{X})^2$
1	x_1	$x_1 - \bar{x}$	$(x_1 - \bar{x})^2$
2	x_2	$x_2 - \bar{x}$	$(x_2 - \bar{x})^2$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	x_n	$x_n - \bar{x}$	$(x_n - \bar{x})^2$
			$\sum_{i=1}^n (x_i - \bar{x})^2$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

3.2.3 Calculation of variance - method 2

i	X	X^2
1	x_1	x_1^2
2	x_2	x_2^2
$\vdots$	$\vdots$	$\vdots$
n	x_n	x_n^2
	$\sum_{i=1}^n x_i$	$\sum_{i=1}^n x_i^2$

$$s^2 = \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2 \right]$$

3.2.4 Formula for variance - proof

$$\begin{aligned}
s^2 &= \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \\
&= \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x}) \\
&= \frac{1}{n-1} \sum_{i=1}^n (x_i^2 - x_i \bar{x} - \bar{x} x_i + \bar{x}^2) \\
&= \frac{1}{n-1} \sum_{i=1}^n (x_i^2 - 2\bar{x} x_i + \bar{x}^2) \\
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - \sum_{i=1}^n 2\bar{x} x_i + \sum_{i=1}^n \bar{x}^2 \right] \\
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + n\bar{x}^2 \right] \\
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - 2\bar{x} (n\bar{x}) + n\bar{x}^2 \right] \\
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \right] \\
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right] \\
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - n \left(\frac{1}{n} \sum_{i=1}^n x_i \right)^2 \right] \\
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - n \left(\frac{1}{n^2} \left(\sum_{i=1}^n x_i \right)^2 \right) \right] \\
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - \frac{n}{n^2} \left(\sum_{i=1}^n x_i \right)^2 \right] \\
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2 \right]
\end{aligned}$$

$$\sum cX = c \sum X \longrightarrow c = 2\bar{x}$$

$$\sum c = nc \longrightarrow c = \bar{x}^2$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$n\bar{x} = \sum_{i=1}^n x_i$$

Excel:

$$s^2 = \text{VAR.S}(\text{data})$$

$$s = \text{STDEV.S}(\text{data})$$

CASIO:

s : SHIFT -> 1 -> 4:Var -> 4: sx

3.2.5 Population variance and standard deviation

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

$$\sigma^2 = \frac{1}{N} \left[\sum_{i=1}^N x_i^2 - \frac{1}{N} \left(\sum_{i=1}^N x_i \right)^2 \right]$$

$$\sigma = \sqrt{\sigma^2}$$

Excel:

$$\sigma^2 = \text{VAR.P}(\text{data})$$

$$\sigma = \text{STDEV.P}(\text{data})$$

CASIO:

σ : SHIFT -> 1 -> 4:Var -> 3: σx

3.3 Coefficient of variation (CV)

$$CV = \frac{s}{\bar{x}} \times 100$$

- relative variation (%)
- relativity makes comparisons possible
 - compare variation relative to mean

4 Measures of shape

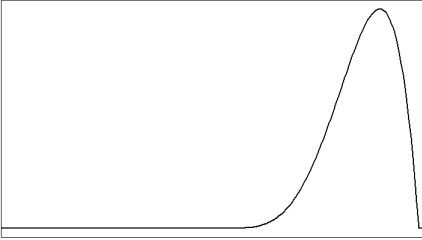
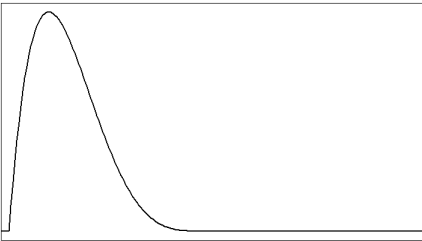
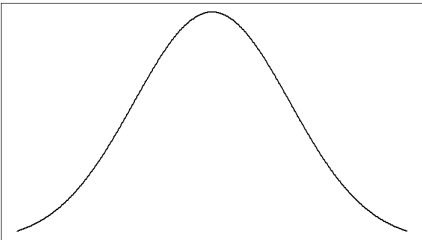
Shape: pattern of distribution

Example of question: "Describe the shape of the data"

4.1 Skewness

Formulas are not covered in SDS188

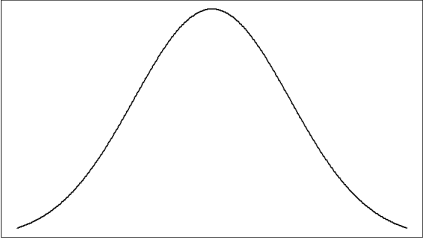
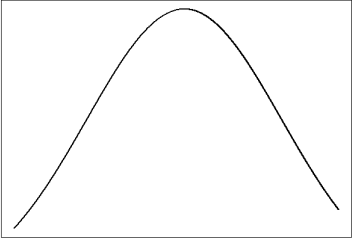
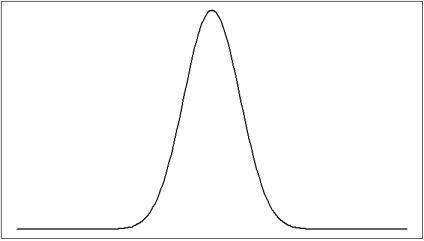
Excel: =SKEW(data)

Data is distributed skew to the left if the mean is smaller than the median, $\bar{x} < \text{mediaan}$. Skewness statistic < 0	
Data is distributed skew to the right if the mean is larger than the median, $\bar{x} > \text{mediaan}$. Skewness statistic > 0	
Data is symmetrically distributed if the mean is approximately equal to the median, $\bar{x} \approx \text{mediaan}$. Skewness statistic ≈ 0	

4.2 Kurtosis

Formulas are not covered in SDS188

Excel: =KURT(data)

The data has a bell-shaped distribution. Kurtosis ≈ 0	
The data has a flatter peak (point of concentration). Kurtosis < 0	
The data has a sharper peak (point of concentration). Kurtosis > 0	

5 Frequency and grouped data

5.1 Frequency data

Discrete numerical data

Data:

F_i represents the number of times that x_i occurs

X	F	FX	$F(X^2)$
x_1	F_1	$F_1 \times x_1$	$F_1 \times x_1^2$
x_2	F_2	$F_2 \times x_2$	$F_2 \times x_2^2$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
	$\sum F$	$\sum FX$	$\sum F(X^2)$

$$n = \sum F$$

$$\bar{X} = \frac{1}{n} \sum FX$$

$$s^2 = \frac{1}{n-1} \left[\sum F(X^2) - \frac{1}{n} (\sum FX)^2 \right]$$

5.2 Grouped or interval data

Data:

Groups/classes	f	x	fx	$f(x^2)$
Class 1	f_1	x_1	f_1x_1	$f_1(x_1^2)$
Class 2	f_2	x_2	f_2x_2	$f_2(x_2^2)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
Total	$\sum f$		$\sum fx$	$\sum f(x^2)$

$$n = \sum f$$

$$\bar{X} = \frac{1}{n} \sum fx$$

$$s^2 = \frac{1}{n-1} \left[\sum f(x^2) - \frac{1}{n} (\sum fx)^2 \right]$$

6 Further exploration of data

Excel:

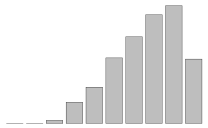
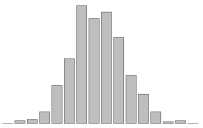
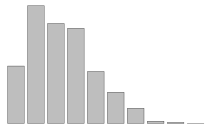
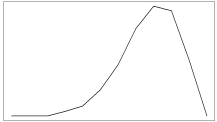
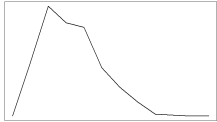
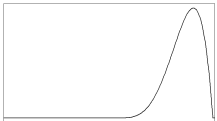
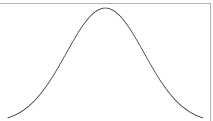
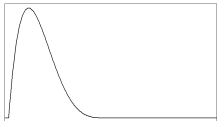
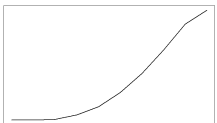
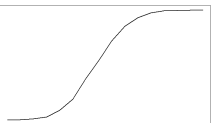
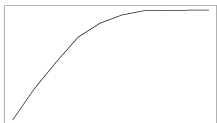
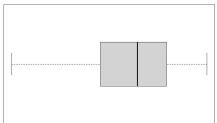
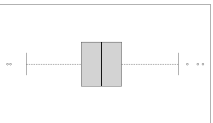
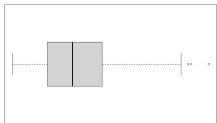
$Min = \text{QUARTILE}(\text{data}, 0)$

$Q_1 = \text{QUARTILE}(\text{data}, 1)$

$Q_2 = \text{QUARTILE}(\text{data}, 2)$

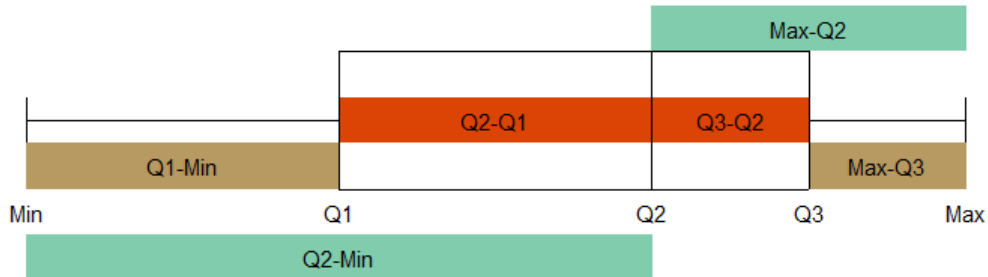
$Q_3 = \text{QUARTILE}(\text{data}, 3)$

$Max = \text{QUARTILE}(\text{data}, 4)$

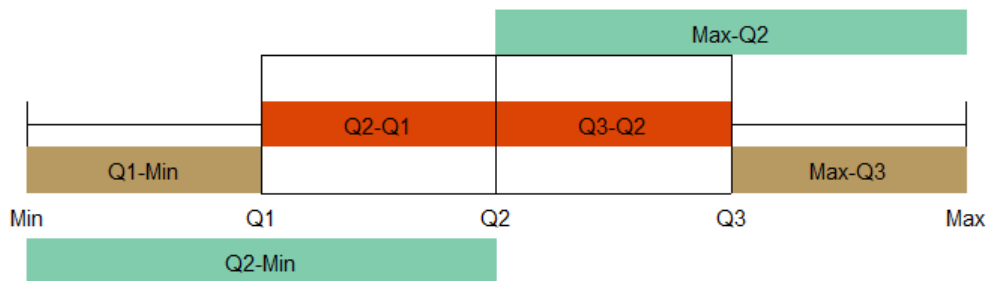
	Skew to the left	Symmetric	Skew to the right
Skewness statistic	< 0	≈ 0	> 0
Central tendency	$\bar{x} < Q_2$	$\bar{x} \approx Q_2$	$\bar{x} > Q_2$
Five number summary	$(Q_2 - \text{Min}) > (\text{Max} - Q_2)$	$(Q_2 - \text{Min}) \approx (\text{Max} - Q_2)$	$(Q_2 - \text{Min}) < (\text{Max} - Q_2)$
	$(Q_1 - \text{Min}) > (\text{Max} - Q_3)$	$(Q_1 - \text{Min}) \approx (\text{Max} - Q_3)$	$(Q_1 - \text{Min}) < (\text{Max} - Q_3)$
	$(Q_2 - Q_1) > (Q_3 - Q_2)$	$(Q_2 - Q_1) \approx (Q_3 - Q_2)$	$(Q_2 - Q_1) < (Q_3 - Q_2)$
Histogram			
Relative frequency polygon			
Normal density (Hfstk 6)			
Ogive			
Boxplot			

6.1 Relationship between five number summary en shape of distribution

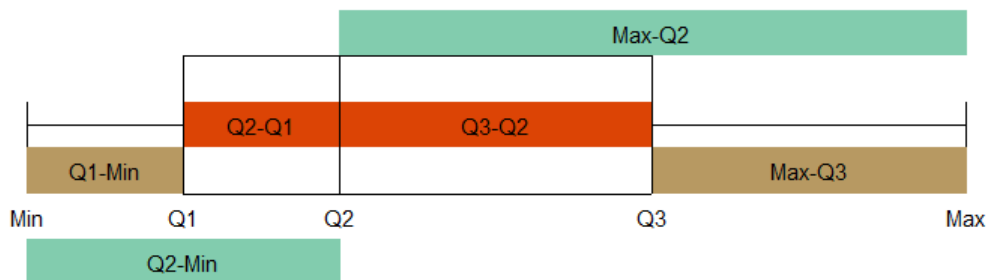
6.1.1 Skew to the left



6.1.2 Symmetric



6.1.3 Skew to the right



7 Relationships between two numerical variables

Measures that quantify the relationship between two variables

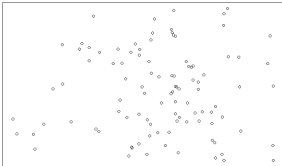
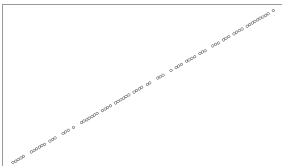
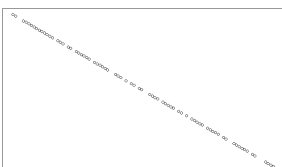
7.1 Covariance

Sample:

$$\text{cov}(X, Y) = s_{xy}$$

Population:

$$\text{cov}(X, Y) = \sigma_{xy}$$

$s_{xy} = 0$	No linear relationship. X and Y are independent.	
$s_{xy} > 0$	Positive linear relationship. An increase in X results in an increase in Y.	
$s_{xy} < 0$	Negative linear relationship. An increase in X results in a decrease in Y.	

7.1.1 Formulas for covariance

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$
$$s_{xy} = \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - \frac{1}{n} \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right) \right]$$
$$\sigma_{xy} = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_X)(y_i - \mu_Y)$$

Excel:

s_{xy} =COVARIANCE.S(dataX,dataY)

σ_{xy} =COVARIANCE.P(dataX,dataY)

7.1.2 Formulas for covariance - proof

$$\begin{aligned} s_{xy} &= \frac{1}{n-1} \sum_{i=1}^n [(x_i - \bar{x})(y_i - \bar{y})] \\ &= \frac{1}{n-1} \sum_{i=1}^n [x_i y_i - x_i \bar{y} - y_i \bar{x} + \bar{x} \bar{y}] \\ &= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \bar{y} - \sum_{i=1}^n y_i \bar{x} + \sum_{i=1}^n \bar{x} \bar{y} \right] \\ &= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - \bar{y} \sum_{i=1}^n x_i - \bar{x} \sum_{i=1}^n y_i + n \bar{x} \bar{y} \right] \\ &= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - \bar{y}(n\bar{x}) - \bar{x}(n\bar{y}) + n\bar{x}\bar{y} \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y} - n\bar{x}\bar{y} + n\bar{x}\bar{y} \right] \\
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y} \right] \\
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - n \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \left(\frac{1}{n} \sum_{i=1}^n y_i \right) \right] \\
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - n \left(\frac{1}{n^2} \sum_{i=1}^n x_i \sum_{i=1}^n y_i \right) \right] \\
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - \frac{n}{n^2} \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right) \right] \\
&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - \frac{1}{n} \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right) \right]
\end{aligned}$$

7.2 Correlation coefficient

Sample correlation coefficient:

$$r$$

Population correlation coefficient:

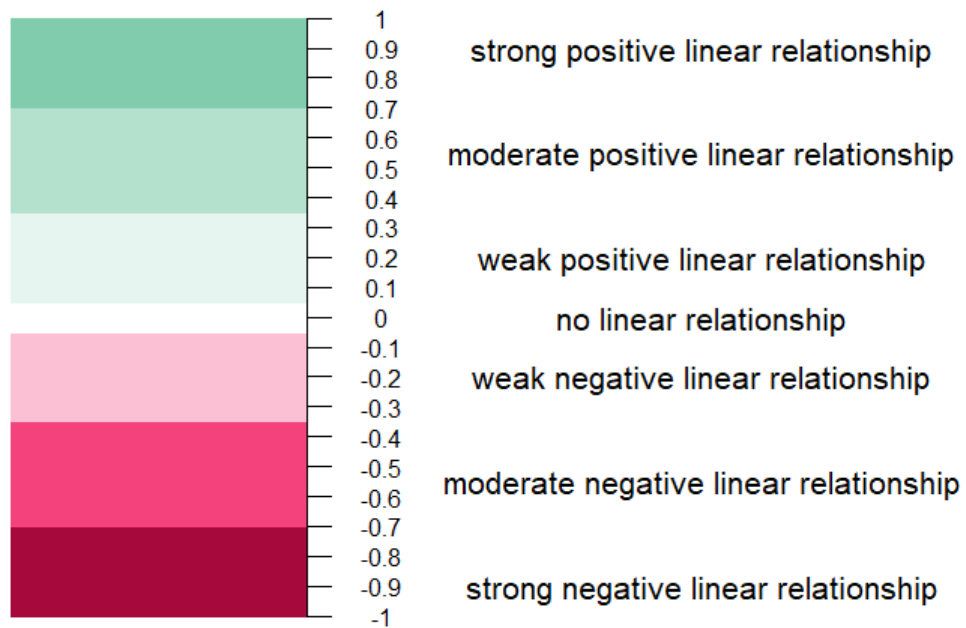
$$\rho$$

7.2.1 Formulas for correlation coefficient

$$\begin{aligned}
r &= \frac{s_{xy}}{s_x s_y} \\
\rho &= \frac{\sigma_{xy}}{\sigma_x \sigma_y}
\end{aligned}$$

Excel: =CORREL(data)

7.2.2 Strength of linear relationship



8 Questions

8.1 Question 1

The heights of 18-year old students have a bell-shaped distribution with a mean of 176cm and a standard deviation of 3.5cm.

1. What is the proportion of students that are between 172.5cm and 179.5cm tall?
2. Which interval centred around the mean contains 95% of the students?

8.2 Question 2

How does one interpret the first quartile, median and third quartile?

The first quartile (Q_1) splits the smallest _____ of the values from the other larger _____. The second quartile (Q_2), splits the data such that _____ of the values are smaller than the median and _____ larger or equal to the median. The third quartile (Q_3) splits the smallest _____ of the values from the largest _____.

8.3 Question 3

Exercise 3.46 in textbook

X	Y
80	6
100	2
100	4
110	4
130	4
190	11
200	10

1. Calculate the covariance.
2. Calculate the correlation coefficient.

3. Which measure is more valuable in expressing the relationship between X and Y ? Why?
4. What conclusion can be made about the relationship between X and Y ?

8.4 Question 4

Data: number of calories per portion of bowls of cereal

80	100	100	110	130	190	200
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1. Calculate the range.
2. Interpret the range.

8.5 Question 5

The following values represent an investment's value over 5 years. Calculate the geometric mean rate of return.

X_1	X_2	X_3	X_4	X_5
1000	1050	1155	1386	693

8.6 Question 6

$$X = 3, 1, 5, 3, 2, 4, 2, 5, 1, 2, 2, 2, 5, 3, 2$$

1. Approximate the mean.
2. Approximate the standard deviation.

8.7 Question 7

A sample of size $n = 50$ has a mean of 28 and a standard deviation of 3. Given this information alone, what can be deduced about the number of observations in the interval $(22; 34)$? Hint: Calculate k and then the proportion

8.8 Question 8

Wat meet die Z waarde?

8.9 Question 9

Answer the following questions about the following data.

Data: Parking spaces for each house in Hampton street

Let X = number of parking spaces per house

F = number of houses with X parking spaces

X	F
1	15
2	27
3	8
4	5

1. Approximate the mean number of parking spaces.
2. Approximate the standard deviation of the number of parking spaces.

8.10 Question 10

How does the empirical rule differ from the Chebyshev rule?

The empirical rule applies to a _____ distribution and the Chebyshev rule applies to _____ distribution.

8.11 Question 11

Data: lengths of leaves on oak tree (in cm)

9, 16, 13, 7, 8, 4, 18, 10, 17, 18, 9, 12, 5, 9, 9, 16, 1, 8, 17, 1, 10, 5, 9, 11, 15, 6, 14, 9, 1, 12, 5, 16, 4, 16, 8, 15, 14, 17

1. Use five classes and construct a frequency distribution for the lengths. [Only the frequency and midpoint columns are required]
2. Use the frequency distribution to approximate the mean and standard deviation of the lengths.